CUSTOMER SENTIMENT ANALYSIS ON FLIPKART PRODUCT REVIEWS

B.Tech Data Analytics Major Project Report

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1. Software Requirements Specification (SRS)

1.1 Introduction

This section outlines the software, logical, and structural requirements for the Flipkart Customer Sentiment Analytics pipeline. The system processes text-based customer evaluations along with categorical numerical ratings to produce clear sentiment classifications and aggregate data points for business intelligence insights.

1.2 Functional Requirements

- **Data Ingestion Engine (Stage 1):** The system must accept local tabular `.csv` data structures containing `Product_name` (string), `Review` (string), and `Rating` (integer).
- **NLP Tokenization & Cleaning Pipeline:** Automated removal of HTML tags, punctuation, special symbols, and specialized e-commerce domain specific stopwords (e.g., 'laptop', 'flipkart', 'product').
- **Advanced Sentiment Analytics Architecture:** Integration of lexicon-based Natural Language Processing to extract numerical Polarity (-1.0 to +1.0) and Subjectivity (0.0 to 1.0) scores.
- **8-Pane Dashboard Visualizations:** Generation of a cohesive multi-pane canvas including pie distribution, count metrics, polarity box plots, density curves, multi-variable heatmaps, and keyword word clouds.

1.3 Non-Functional Requirements

- **Performance:** Process batch operations of over 2,300 text records in under 3.5 seconds entirely offline without network latency.
- **Accuracy Validation:** Statistical text polarity scores must match and correlate effectively with actual customer numerical star ratings to ensure consistency.

1.4 Data Dictionary & Target Schema

Field Column Identifier	Data Type	Constraints / Bounds	Description
Product_name	String (object)	Non-Null Row Values	The name/variant of the consumer electronics or laptop model.
Review	String (object)	Non-Null Text	Raw literal feedback text written by the purchasing customer.
Rating	Integer (int64)	Bounded values: 1 to 5	Quantitative scale metric representing customer satisfaction.
Polarity	Float (float64)	Continuous: -1.0 to +1.0	Calculated sentiment metric tone score.

Field Column Identifier	Data Type	Constraints / Bounds	Description
Sentiment	String (object)	Positive, Negative, Neutral	Categorical bucket determined via text classification.

2. Data Processing Pipeline & Evaluation Metrics

2.1 Stage 1 Ingestion Profiling

Upon file validation, the system reads exactly 2,304 clean records. The index and unneeded columns are dropped programmatically to keep memory consumption low (~54.1 KB), providing a scalable foundation for execution.

Macro Sentiment Distribution Count Summary

During automated tokenization, the dataset was discovered to possess a strong consumer satisfaction bias. The absolute classifications yield:

- **Positive Sentiment Buckets:** 2,062 individual reviews
- **Negative Sentiment Buckets:** 156 individual reviews
- **Neutral Sentiment Buckets:** 86 individual reviews

2.2 Targeted Product Analytics Insights

By implementing a multi-key group aggregation pattern (`df.groupby('Product_name')`), the application successfully separates baseline numbers from actionable business intelligence. The system automatically isolated the top 5 product items with the lowest emotional polarity thresholds among items with meaningful review samples ($N \geq 5$):

Product Naming Model Identifier	Avg User Rating	Avg Polarity Score	Sample Size
realme C21Y (Cross Black, 32 GB) (3 GB RAM)	3.10 ★	**-0.3933**	20 reviews
realme C21Y (Cross Blue, 32 GB) (3 GB RAM)	3.10 ★	**-0.3933**	20 reviews
Redmi 9A (Sea Blue, 32 GB) (3 GB RAM)	3.10 ★	**-0.1241**	10 reviews
Redmi 9A (Midnight Black, 32 GB) (3 GB RAM)	3.10 ★	**-0.1241**	10 reviews
Redmi 9A (Nature Green, 32 GB) (3 GB RAM)	3.10 ★	**-0.1241**	10 reviews

2.3 Summary of Visual Analytical Dashboard Features

The system visualizes its results via an advanced, custom 8-pane plotting canvas. It includes custom **Pie Charts** to track sentiment distribution percentages, **Seaborn Box Plots** to check for outliers and rating alignment, a continuous **KDE Density Curve** to view the text polarity distribution, and a **Correlation Matrix Heatmap** to study relationships between metadata features like review length and subjectivity.